

MTU_Value*Service* **Technical Documentation**

Diesel Engine
V 396 TB94
Application Group 1DS

Maintenance Schedule
M050706/05E



applicable for ...

12V396TB94
16V396TB94

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Subject to alterations and amendments.

1 Preface

MTU maintenance concept

The maintenance system for MTU products is based on a preventive maintenance concept. Preventive maintenance facilitates advance planning and ensures a high level of equipment availability.

The maintenance schedule is based on the load profile / load factor specified below. The time intervals at which the maintenance work is to be carried out and the relevant checks and tasks involved are average values based on operational experience and are therefore to be regarded as guidelines only. Special operating conditions and technical requirements may require additional maintenance work and/or modification of the maintenance intervals. In order to be authorized to carry out the individual maintenance jobs, maintenance personnel must have achieved a level of training and qualification appropriate to the complexity of the task in hand. The various Qualification Levels QL1 to QL4 reflect the levels of training offered in MTU courses and the contents of the tool kits required:

- QL1: Operational monitoring and maintenance which can be carried out during a break in operation without disassembling the engine.
- QL2: Component exchange (corrective only).
- QL3: Maintenance work which requires partial disassembly of the engine.
- QL4: Maintenance work which requires complete disassembly of the engine.

The maintenance schedule matrix normally finishes with extended component maintenance. Following this, maintenance work is to be continued at the intervals indicated.

The numbers stated in the list of jobs provide a reference to the scope of parts required.

Notes on maintenance

Specifications for fluids and lubricants, guideline values for their maintenance and change intervals and lists of recommended fluids and lubricants are contained in the MTU Fluids and Lubricants Specifications A001061 and in the fluids and lubricants specifications produced by the component manufacturers. They are therefore not included in the maintenance schedule (exception: deviations from the Fluids and Lubricants Specifications). All fluids and lubricants used must meet MTU specifications and be approved by the relevant component manufacturer.

Amongst other items, the operator/customer must carry out the following additional maintenance work:

- Protect components made of rubber or synthetic material from oil. Never treat them with organic detergents. Wipe with a dry cloth only.
- Fuel prefilter:
The maintenance interval depends on how dirty the fuel is. The paper inserts in fuel prefilters must be changed every two years at the latest (Task 9998).
- Battery:
Battery maintenance depends on the level of use and the ambient conditions. The battery manufacturer's instructions must be obeyed.

The relevant manufacturer's instructions apply with respect to the maintenance of any components which do not appear in this maintenance schedule.

This Maintenance Schedule may include components which are not installed on your engine; these may be disregarded.

Out-of-service periods

If the engine is to remain out of service for more than 1 month, carry out engine preservation procedures in accordance with the Fluids and Lubricants Specifications, MTU Publication No. A001061.

Application Group

1DS Continuous operation, variable, low load

Load profile

Load factor	100%	70%	<10%
Corresponding operating time	10%	70%	20%

2 Maintenance schedule matrix

0-5.250 Operating hours

Item	Limit	Operating hours [h]																						
		Daily	250	500	750	1.000	1.250	1.500	1.750	2.000	2.250	2.500	2.750	3.000	3.250	3.500	3.750	4.000	4.250	4.500	4.750	5.000	5.250	
Engine operation	-	X																						
Engine oil filter	2 a																							
Emergency air shut-off flaps	1 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Engine oil edge-type filter	1 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Centrifugal oil filter	1 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	
Valve gear	18 a			X		X		X		X		X		X		X		X		X		X		
Air filters	1 a			X		X		X		X		X		X		X		X		X		X		
Lubrication points	1 a			X		X		X		X		X		X		X		X		X		X		
Fuel filter	2 a			X		X		X		X		X		X		X		X		X		X		
Starter	18 a							X						X						X				
Fuel injectors	18 a							X						X						X				
Charge-air preheating	18 a							X						X						X				
Battery charging generator	18 a							X						X						X				
Engine coolant circuit	18 a							X						X						X				
Combustion chambers	18 a							X						X						X				
Torsional resilient coupling	2 a							X						X						X				
Engine mounts	2 a							X						X						X				
Pedestal bearing	2 a							X						X						X				
Air filters	3 a							X						X						X				
Turbochargers	18 a													X										
Turbocharger control system	18 a													X										
Component maintenance	18 a													X										
Component maintenance	18 a													X										
Battery charging generator	18 a													X										
Gear coupling	2 a													X										
Cylinder heads	18 a																							
Extended component maintenance	18 a																							
w = weeks m = months a = years																								

5.500-6.000 Operating hours

Item	Limit	Operating hours [h]																			
		5.500	5.750	6.000																	
Engine operation	-																				
Engine oil filter	2 a																				
Emergency air shut-off flaps	1 a	X	X	X																	
Engine oil edge-type filter	1 a	X	X	X																	
Centrifugal oil filter	1 a	X	X	X																	
Valve gear	18 a	X		X																	
Air filters	1 a	X		X																	
Lubrication points	1 a	X		X																	
Fuel filter	2 a	X		X																	
Starter	18 a			X																	
Fuel injectors	18 a			X																	
Charge-air preheating	18 a			X																	
Battery charging generator	18 a			X																	
Engine coolant circuit	18 a			X																	
Combustion chambers	18 a			X																	
Torsional resilient coupling	2 a			X																	
Engine mounts	2 a			X																	
Pedestal bearing	2 a			X																	
Air filters	3 a			X																	
Turbochargers	18 a			X																	
Turbocharger control system	18 a			X																	
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